

MIT Art Design and Technology University
MIT School of Computing, Pune



MIT-ADT
UNIVERSITY
PUNE, INDIA

Department of Computer Science and Engineering

23CSE3154- Azure & GCP Essential Services

Class - TY. (SEM-VI)

Unit - V

AI and Cloud Computing

AY 2025-2026 SEM-II





Text Books

GCP: Google Cloud Platform: Learn Google Cloud Platform from the Scratch: The Ultimate Guide for Beginners by Adney Ainsley

Google Cloud Platform from the very beginning Kindle Edition by Jerry N. P.





Reference Books

Enterprise Cloud Computing - Technology, Architecture, Applications, Gautam Shroff,
Cambridge University Press, 2010

Cloud Security: A Comprehensive Guide to Secure Cloud Computing, Ronald L. Krutz,
Russell Dean Vines, Wiley- India, 2010





Unit V–AI and Cloud Computing

Introduction to AI and Cloud Computing Advantages and Challenges AI and Cloud Computing, Cloud Service Models for AI, AI Cloud Architecture, AI Infrastructure in Cloud, Serverless AI, Use Cases of AI with Cloud Computing like Healthcare (disease detection, medical imaging), Agriculture (crop disease detection).





1. Introduction to AI and Cloud Computing

- ♦ **Artificial Intelligence (AI)**

Artificial Intelligence is a branch of computer science that enables machines to perform tasks that normally require human intelligence such as learning, reasoning, problem-solving, and decision-making.

Examples: speech recognition, image classification, chatbots.

- ♦ **Cloud Computing**

Cloud computing is the delivery of computing services (servers, storage, databases, networking, software) over the internet. It allows users to access resources on demand without owning physical hardware.

- ♦ **AI + Cloud Integration**

Cloud platforms provide the infrastructure (storage + computing power) needed to build, train, and deploy AI models efficiently.

👉 AI needs large data and compute → Cloud provides scalability and flexibility.





Advantages and Challenges of AI & Cloud Computing

Advantages

- **Scalability:** Easily increase/decrease computing resources.

In cloud computing, resources are not fixed. If demand increases (e.g., more users or more data processing), the system can automatically **scale up** by adding more resources. When demand decreases, it can **scale down** to save cost.

Types of Scalability

- **Vertical Scaling (Scale Up/Down):**
Increasing or decreasing the power of a single machine (e.g., adding more RAM or CPU).
- **Horizontal Scaling (Scale Out/In):**
Adding or removing multiple machines/servers to distribute the load.

In AI Context

- During **model training**, large datasets require more computing power → scale up.
- During **low usage**, resources are reduced → cost savings.





- **Cost Efficiency:** No need to invest in expensive hardware.

In traditional systems, organizations must buy and maintain servers, storage devices, and networking equipment, which requires **high upfront investment** and maintenance costs. With cloud computing, users can **rent resources on demand** and pay only for what they use (**pay-as-you-go model**).

- **Accessibility:** Access AI services from anywhere.

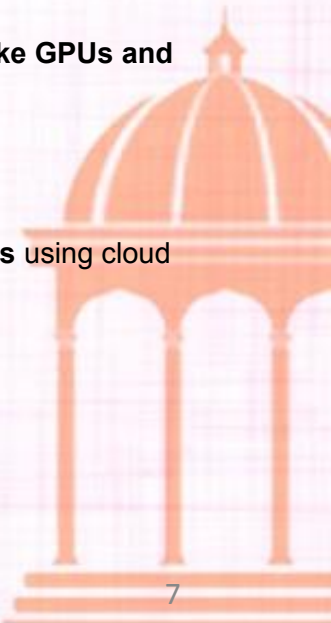
Cloud computing allows users to connect to AI tools and services through web browsers or APIs without being physically present at a specific location. As long as there is an internet connection, users can access resources globally.

- **High Performance:** Cloud provides GPUs/TPUs for faster AI processing.

High performance refers to the ability of cloud platforms to provide **powerful computing resources (like GPUs and TPUs)** that enable faster and efficient processing of AI tasks.

- **Faster Deployment:** Quick development and deployment of AI models.

Faster deployment means the ability to **quickly develop, test, and deploy AI models and applications** using cloud platforms.





Challenges

- **Data Security & Privacy:** Sensitive data stored on cloud may be at risk.

Data security and privacy refer to the protection of **sensitive data (personal, financial, medical, etc.)** stored and processed in cloud-based AI systems from unauthorized access, misuse, or breaches.

- **Latency Issues:** Delay due to network dependency. **Latency** refers to the **delay or time taken for data to travel between the user and the cloud server**. In cloud-based AI systems, latency occurs due to dependence on network communication.
- **Dependence on Internet:** Requires stable connectivity. **Dependence on internet** means that cloud-based AI systems require a **stable and continuous internet connection** to function properly.
- **Compliance Issues:** Must follow regulations (especially in healthcare). **Compliance issues** refer to the need for organizations to **follow legal, regulatory, and industry standards** when using AI and cloud computing, especially when handling sensitive data.
- **Cost Management:** Improper usage can increase cloud costs. **Cost management** refers to the process of **monitoring, controlling, and optimizing cloud expenses** to avoid unnecessary spending.



Cloud Service Models for AI

• IaaS (Infrastructure as a Service)

- Provides virtual machines, storage, and networking.
- Users control OS and applications.
- Example use: Training custom AI models with full control.

Definition

IaaS provides **basic computing resources** such as virtual machines, storage, and networking over the cloud.

Features

- Full control over operating system and software
- Flexible and scalable infrastructure
- User manages applications and data

In AI Context

- Used for **training complex AI models**
- Developers install frameworks like TensorFlow, PyTorch
- Suitable for deep learning projects requiring GPUs

Example Use

A company trains a custom AI model using cloud-based GPU virtual machines.





♦ PaaS (Platform as a Service)

- Provides platform to develop and deploy AI apps.
- No need to manage infrastructure.
- Example use: Building ML models using cloud tools.

📌 Definition

PaaS provides a **platform and environment** to develop, test, and deploy applications without managing underlying infrastructure.

⚙️ Features

- No need to manage servers or OS
- Built-in development tools and AI frameworks
- Faster development and deployment

🤖 In AI Context

- Used to build and deploy ML models easily
- Includes tools for data processing, training, and testing

✅ Example Use

A developer builds a machine learning model using a cloud ML platform without worrying about infrastructure.





♦ SaaS (Software as a Service)

- Ready-to-use AI applications.
- No need to develop or manage anything.
- Example use: AI APIs (chatbots, image recognition).

Definition

SaaS provides **ready-to-use software applications** over the internet.

Features

- No installation or maintenance required
- Accessible via web browser or API
- Fully managed by provider

In AI Context

- Pre-built AI services like chatbots, speech recognition, image analysis
- No need to build models from scratch

Example Use

Using an AI-powered image recognition API directly in an application.





♦ FaaS (Function as a Service / Serverless)

- Runs code in response to events.
- No server management required.
- Example: AI inference triggered by API call.

📌 Definition

FaaS allows developers to run **small pieces of code (functions)** in response to events without managing servers.

⚙️ Features

- Event-driven execution
- Automatic scaling
- Pay only for execution time

🤖 In AI Context

- Used for **AI inference tasks** (e.g., prediction when user uploads image)
- Ideal for lightweight and real-time AI operations

✅ Example Use

Running an AI prediction function when a user submits data on a website.





Comparison Table

Feature	IaaS	PaaS	SaaS	FaaS (Serverless)
Full Form	Infrastructure as a Service	Platform as a Service	Software as a Service	Function as a Service
User Control	High	Medium	Low	Very Low
Managed By Provider	Hardware only	Hardware + OS + Runtime	Everything	Everything (including execution)
User Responsibility	OS, apps, data	Apps & data	Only usage	Only function code
Setup Time	High	Medium	Low	Very Low
Flexibility	Very High	Moderate	Low	Limited
Scalability	Manual/Auto	Auto	Auto	Fully automatic
AI Usage	Custom model training	Model development & deployment	Pre-built AI tools	Event-based AI inference
Cost	Pay for infrastructure	Pay for platform	Subscription-based	Pay per execution



AI Cloud Architecture

1. Data Layer

- Stores structured/unstructured data.
- Includes databases and data lakes. architecture consists of multiple layers:

 **Definition:-**The **Data Layer** is the foundation of AI cloud architecture that is responsible for **storing, managing, and organizing data** required for AI applications.

Types of Data Stored

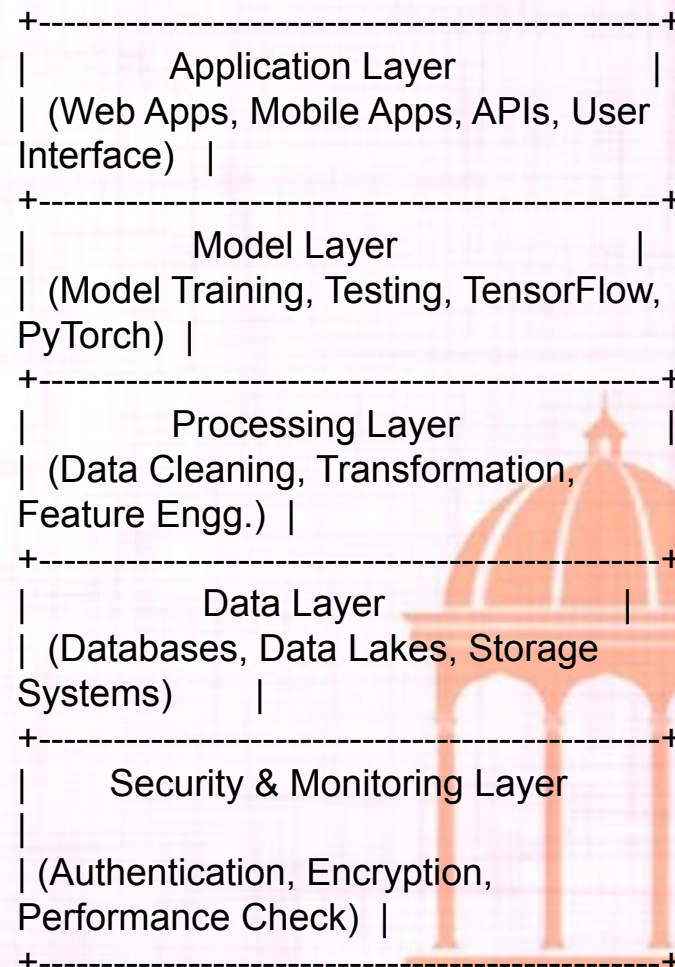
- **Structured Data** → Organized data (tables, databases, rows & columns)
Example: customer records, transaction data
- **Unstructured Data** → Raw data without fixed format
Example: images, videos, audio, text files.

Components of Data Layer

- **Databases** → Store structured data (SQL, NoSQL)
- **Data Lakes** → Store large volumes of raw/unstructured data
- **Data Warehouses** → Store processed and analyzed data
- **Cloud Storage** → Scalable storage for AI datasets

Role in AI Systems

- Provides **training data** for AI models
- Stores **real-time and historical data**
- Ensures data is **available, reliable, and scalable**
- Supports **data preprocessing and analytics**





♦ 2. Processing Layer

- Data cleaning and preprocessing.
- Uses distributed computing systems.

Definition

The **Processing Layer** is responsible for **cleaning, transforming, and preparing raw data** so it can be used effectively for AI model training and analysis.

Key Functions

- **Data Cleaning** → Removing errors, duplicates, and missing values
- **Data Transformation** → Converting data into suitable formats
- **Data Integration** → Combining data from multiple sources
- **Feature Engineering** → Selecting important variables for AI models

Technologies Used

- **Distributed Computing Systems** → Process large datasets across multiple machines
- **Parallel Processing** → Perform multiple operations simultaneously
- Tools like data processing frameworks (e.g., batch and stream processing systems)

Role in AI Systems

- Converts raw data into **high-quality input** for AI models
- Improves **accuracy and efficiency** of models
- Handles **large-scale data processing** in cloud environments
-





♦ 3. Model Layer

- Training and testing AI/ML models.
- Uses frameworks like TensorFlow, PyTorch.
-  **Definition**

The **Model Layer** is responsible for **building, training, testing, and optimizing AI/ML models** using processed data.

Key Functions

- **Model Training** → Learning patterns from data
- **Model Testing & Validation** → Evaluating accuracy and performance
- **Hyperparameter Tuning** → Improving model efficiency
- **Model Optimization** → Reducing errors and improving speed

Tools & Frameworks

- **TensorFlow** → Popular for deep learning models
- **PyTorch** → Flexible framework for research and development
- Other ML libraries for training and evaluation

Role in AI Systems

- Converts processed data into **intelligent models**
- Learns patterns, relationships, and predictions
- Acts as the **core intelligence** of the AI system





◆ 4. Application Layer

- Deploys AI models as applications or APIs.
- End-users interact with this layer.

Definition

The **Application Layer** is the top layer of AI cloud architecture where **trained AI models are deployed as applications or APIs**, and **end-users interact with the system**.

Key Functions

- **Model Deployment** → Making trained models available for use
- **API Integration** → Connecting AI models with applications
- **User Interface (UI)** → Providing dashboards, apps, or web interfaces
- **Real-time Inference** → Generating predictions based on user input

Components

- Web applications
- Mobile applications
- REST APIs / AI services
- User dashboards



Role in AI Systems

- Acts as a **bridge between users and AI models**
- Delivers AI functionality in a usable form





◆ 5. Security & Monitoring Layer

- Ensures data protection, authentication, and performance monitoring.

Definition

The **Security & Monitoring Layer** is responsible for **protecting data and AI systems** and continuously **monitoring system performance, usage, and reliability**.

Security Functions

- **Data Protection** → Safeguards sensitive data from unauthorized access
- **Authentication** → Verifies user identity (login systems, multi-factor authentication)
- **Authorization** → Controls access permissions
- **Encryption** → Protects data during storage and transmission

Monitoring Functions

- **Performance Monitoring** → Tracks system speed and efficiency
- **Resource Monitoring** → Observes CPU, memory, and storage usage
- **Error Detection** → Identifies failures or issues in AI models
- **Logging & Auditing** → Keeps records of system activities

Role in AI Systems

- Ensures **secure handling of AI data and models**
- Maintains **system reliability and uptime**
- Detects anomalies and prevents failures





AI Infrastructure in Cloud

AI infrastructure includes hardware and software components required to run AI systems:

◆ Compute Resources

Definition

Compute resources refer to the **processing power provided by cloud systems** (such as CPUs, GPUs, and TPUs) used to **train and run AI/ML models**

Types of Compute Resources

◆ 1. CPU (Central Processing Unit)

- General-purpose processor
- Suitable for basic tasks and small-scale AI workloads
- Slower for heavy computations

◆ 2. GPU (Graphics Processing Unit)

- Designed for **parallel processing**
- Ideal for **deep learning and large datasets**
- Faster than CPUs for AI training

◆ 3. TPU (Tensor Processing Unit)

- Specialized hardware designed for AI by Google
- Optimized for machine learning tasks
- Provides very high performance for training models





- ◆ Storage

Definition

Storage in AI cloud infrastructure refers to systems used to **store, manage, and access large volumes of data** required for AI/ML models.

Types of Storage

◆ 1. Object Storage

- Used for **large-scale unstructured data**
- Stores data as objects (images, videos, audio, files)
- Highly scalable and cost-effective

👉 *Example:* Storing datasets for image recognition or medical imaging

◆ 2. Databases (Structured Storage)

- Used for **structured data** (tables, rows, columns)
- Supports fast querying and transactions
- Includes SQL and NoSQL databases

👉 *Example:* Customer records, financial data

Role in AI Systems

- Stores **training datasets** and **model outputs**
- Maintains both **raw and processed data**
- Enables quick access to data for training and inference





◆ Networking

Definition

Networking refers to the system of **connecting cloud resources, servers, and users** to enable **fast and secure data transfer** in AI applications.

Key Functions

- **Data Transfer** → Moves data between storage, compute, and users
- **Communication** → Connects different components of AI systems
- **Remote Access** → Allows users to access cloud services from anywhere

Key Features

- **High-speed connectivity** → Ensures fast processing of large datasets
- **Low latency** → Reduces delay in real-time AI applications
- **Secure communication** → Uses firewalls, VPNs, encryption
- **Scalability** → Handles increasing network traffic

Role in AI Systems

- Transfers large datasets for **model training**
- Enables **real-time AI services** (e.g., chatbots, healthcare monitoring)
- Supports **distributed computing** across multiple servers





♦ Frameworks & Tools

- **AI Libraries:**

Popular libraries like TensorFlow and PyTorch are used to build, train, and test machine learning and deep learning models.

- **Model Deployment Tools:**

Tools such as Docker and Kubernetes help deploy models into production environments efficiently.

- **Monitoring Tools:**

Platforms like Prometheus and Grafana are used to monitor model performance, system health, and detect issues in real time.





♦ Distributed Systems

- Enable parallel processing of large datasets.
- Distributed systems enable **parallel processing** of large datasets by dividing tasks across multiple machines. Frameworks like Apache Hadoop and Apache Spark are commonly used to handle big data efficiently. They improve **scalability**, **speed**, and **fault tolerance** in data processing tasks





6. Serverless AI

♦ Definition

Serverless AI is a cloud computing model where developers build and run AI applications without managing servers.

♦ Features

- Automatic scaling
- Pay-per-use billing
- Event-driven execution

♦ Advantages

- Reduces operational complexity
- Cost-effective for small workloads
- Faster deployment

♦ Limitations

- Cold start latency
- Limited execution time
- Less control over infrastructure





Use Case: Healthcare (AI + Cloud)

◆ Disease Detection

AI models analyze patient data to detect diseases like cancer, diabetes, etc.

◆ Medical Imaging

AI processes X-rays, MRIs, CT scans to identify abnormalities.

◆ Benefits

- Faster diagnosis
- Improved accuracy
- Remote healthcare access

◆ Challenges

- Data privacy (sensitive patient data)
- Regulatory compliance
- Need for high accuracy





Use Case: Agriculture (AI + Cloud)

◆ Crop Disease Detection

AI analyzes images of crops to identify diseases early.

◆ Smart Farming

- Weather prediction
- Soil analysis
- Automated irrigation

◆ Benefits

- Increased crop yield
- Reduced losses
- Efficient resource usage

◆ Challenges

- Limited internet in rural areas
- Data availability
- Cost of implementation

